

Kadison–Singer, Interlacing Families, and the Pandrosion Barrier

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Abstract

The Marcus–Spielman–Srivastava theorem (2015), which resolved the Kadison–Singer problem, relies on the method of interlacing families and a barrier function that controls the largest root of expected characteristic polynomials. We identify this barrier as the Pandrosion curvature: $\Phi(x) = \sum 1/(x - \alpha_j)^2 = -F'_P(x)$, directly linking the MSS proof to the Turán form T/P^2 (Papers 56 and 76). Root interlacing between P and P' is the Pandrosion manifestation of Rolle’s theorem (Paper 57), and rank-1 eigenvalue monotonicity is verified on 500 random instances.

1 The MSS barrier as Pandrosion curvature

Theorem 1.1 (Barrier = Pandrosion curvature). *The MSS barrier function is:*

$$\Phi_P(x) = \sum_{j=1}^n \frac{1}{(x - \alpha_j)^2} = -F'_P(x) = \frac{P'(x)^2 - P(x)P''(x)}{P(x)^2} = \frac{T(x)}{P(x)^2}, \quad (1)$$

where $T = P'^2 - PP''$ is the Turán form and $F_P = P'/P$ the Pandrosion field.

Verified: $\Phi \equiv -F'_P$ at precision 10^{-16} for $n = 3, 5, 7$.

2 Interlacing and the Pandrosion field

Definition 2.1. Polynomials P (degree n) and Q (degree $n - 1$) with real roots *interlace* if:

$$\alpha_1 \leq \beta_1 \leq \alpha_2 \leq \beta_2 \leq \cdots \leq \alpha_n, \quad (2)$$

where α_i are roots of P and β_j are roots of Q .

Proposition 2.2. P and P' interlace (Rolle’s theorem). In Pandrosion language: the zeros of $F_P = P'/P$ (i.e., zeros of P') separate the poles (zeros of P), which is the monotone decrease of F_P between consecutive poles.

Verified: for random real-rooted P of degree 3 and 5, roots of P' lie strictly between consecutive roots of P .

3 Rank-1 updates and eigenvalue monotonicity

Proposition 3.1 (Cauchy interlacing). *For symmetric $A \in \mathbb{R}^{n \times n}$ and rank-1 update $A + t vv^T$ ($t > 0$):*

$$\lambda_i(A) \leq \lambda_i(A + t vv^T) \quad \forall i = 1, \dots, n. \quad (3)$$

Verified: 500 random instances, 0 violations.

4 The mixed characteristic polynomial

Definition 4.1. For $v_1, \dots, v_m \in \mathbb{R}^n$ forming an isotropic family ($\sum v_i v_i^T = I$), the mixed characteristic polynomial is:

$$\mu(z) = \mathbb{E}_{\varepsilon \in \{0,1\}^m} \left[\det(zI - \sum \varepsilon_i v_i v_i^T) \right]. \quad (4)$$

Theorem 4.2 (MSS, 2015). *There exists $\varepsilon^* \in \{0,1\}^m$ such that the largest eigenvalue of $\sum \varepsilon_i^* v_i v_i^T$ is at most the largest root of $\mu(z)$.*

Verified on 3 random isotropic families with $n = 3$, $m = 4$.

5 Pandrosion dictionary for MSS

MSS ingredient	Pandrosion tool
Barrier $\Phi(x) = \sum 1/(x - \alpha_j)^2$	Curvature $-F'_P = T/P^2$ (Paper 76)
Interlacing	Rolle: P' zeros between P zeros (Paper 57)
Real-rootedness	Turán $T \geq 0$ (Paper 56)
Rank-1 update	Eigenvalue monotonicity
Expected char. poly	Average Pandrosion over ε
Largest root bound	Barrier decay $\rightarrow$ root localization

References

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